



Sri Lanka Institute of Information Technology

Web-Based Construction Labor Management System

Sprint 0

Information Systems Project Management 2026

Project ID: ISE_WE_0101_32

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Criteria	Marks allocated	Marks given
Problem Specification	5%	
Business Idea	5%	
Solution Outline	10%	
Key Benefits	10%	
Main Objectives	5%	
Sub Objectives	5%	
Project Plan	10%	
Product Vision statement	05%	
Epics	05%	
User Stories	10%	
Initial product Backlog	05%	
Definition of Done	05%	
Hardware & Software requirements	5%	
Budget	5%	
References	5%	
Formatting	5%	
Total Marks (100)	100%	

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1.Introduction

1.1 Problem Specification

J A L Enterprises works in the construction industry where labour management plays a vital role in timely completion of projects and cost control. In the current scenario, various labour-related operations take place manually using paper and spreadsheets. As the company grows with respect to its labour and sites, managing this information has become difficult and cumbersome. There are chances that the information about the labourers may be duplicated or incomplete and the identification of appropriate labourers for assigning to the site may become problematic. There will be difficulties in maintaining accurate records of attendance and hours of labourers and keeping track of their overtime.

This situation may cause problems with regard to calculating wages of the labourers because the payroll department depends upon the attendance and wage information to calculate their payments. There will be delays, extra workload and chances of mistakes in the payment due to this situation. Lack of a centralized system causes problems for the HR, site managers, supervisors and payroll department to have easy access to reliable and updated information. Thus, there is a need for a centralized system for labour management at J A L Enterprises.

1.2 Business Idea

The proposed business idea is to develop a web-based Construction Labour Management System to help construction companies manage their workforce and day-to-day labour-related activities in one place. Construction companies often have a large number of labourers working across different sites, which can make it difficult to keep workforce information accurate and up to date when using paper records and separate spreadsheets. The proposed system will provide a centralized platform that makes it easier to manage this information in a more organized and efficient way.

The system will provide authorized users with a single platform to handle key labour management activities. These will include maintaining labourer profiles, assigning workers to suitable construction sites, recording daily attendance and working hours, managing overtime, and maintaining wage and payment records. Bringing these activities together in one system can

reduce duplicated work and make it easier for information to move between different labour management processes

The labourer management feature will allow authorized users to create, update, view, and manage individual labourer records. Details such as personal information, contact details, job role, skills, experience, and employment information can be stored in a structured format. This will allow management to quickly find relevant worker information and make better decisions when allocating labour.

The Site Allocation and Work Assignment feature will help management assign labourers to construction sites based on their skills, roles, availability, and project requirements. This can make the allocation process more organized and help ensure that suitable workers are assigned to the appropriate sites. It can also reduce the amount of manual coordination required between management and site supervisors.

Attendance and overtime management will allow site supervisors or other authorized users to record daily attendance, working hours, and overtime. Rather than depending on paper attendance sheets or separate spreadsheets, these records can be stored in the system and accessed when needed. Keeping overtime records in the same system can also help improve the accuracy of wage calculations and reduce errors caused by manual record keeping.

The wage and payment management feature will be used to maintain information about labour wages, overtime payments, deductions, and previous payments. Having attendance, overtime, and wage information connected within the same system can make payment calculations easier to manage and provide a clear record of labour related transactions.

Overall, the system is expected to reduce the time and effort spent on routine labour management tasks. It can also improve data accuracy, make information easier to access, and improve coordination between management and construction site supervisors. Since the information will be stored in a centralized database, authorized users will not have to search through multiple paper documents or spreadsheet files to find the information they need.

1.3 Solution Outline

The proposed solution is a web-based Construction Labour Management System that collects and organizes labor management essential functions of J A L Enterprises. This system will give an ideal platform for the user to manage labor information of workers in the fields of construction. The system also enables its user to track attendance and overtime as well as computes payments payable to workers.

The proposed solution will reduce the need for paper based accounting through the provision of info that is centralized, reliable and accessible. Information related to laborers including fundamental data checks, verification of attendance records, computation of wage claims based on approved amount of work performed, etc. will be supported by the system. Role based access will be provided by the system so that each type of user will have the access to perform the functions related to the level of access.

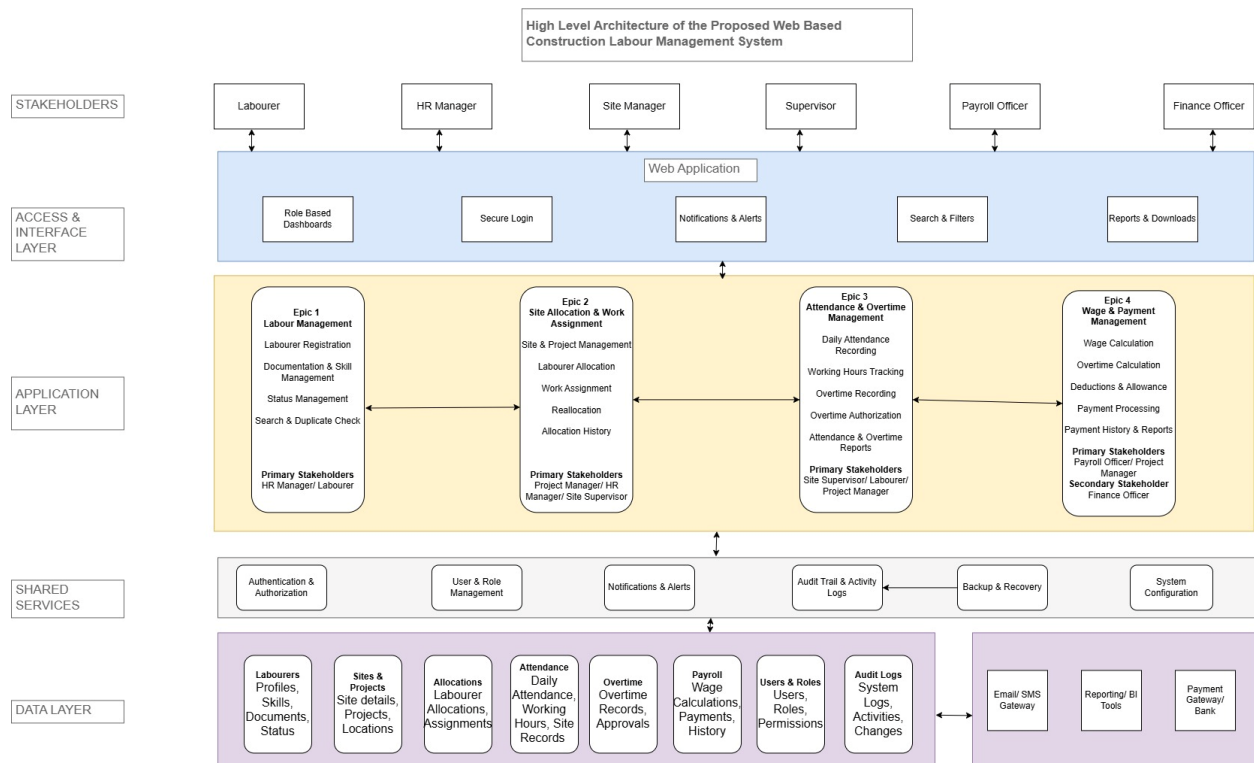


Figure 1.1 High Level Architecture of the Web Based Construction Management System

1.4 Key Benefits

- **Reduced wage calculation errors:** Computing the wages automatically by using the recorded information about attendance, working hours, overtime and corresponding deductions reduce errors in manual computations
- **Improved payment transparency:** Centralized wage and payment information allows authorized personal to track better wages, deductions, payment process and history of payments
- **Reduced administrative workload:** Digital labour records and wage computation processes decrease the efforts needed for managing paper records and performing the repetitive calculations manually
- **Centralized labour information:** Information about worker's characteristics, their attendance and payment history can be kept in one place
- **Faster access to information:** Current information about labour and payment process can be accessed quicker by authorized personal
- **Improved record accuracy and consistency:** Using centralized database helps to avoid duplication of records and increase consistency of the information
- **Better decision making:** Accessible and accurate information about labour and payment process gives management necessary information about cost and other details
- **Improved accountability:** Well managed records about attendance, wages calculations and payments create better historical track of labour transactions and responsibilities

2.Objectives

2.1 Main Objective

To create and develop Web Based Labour Management System that digitalize, keep on track and manage workers attendance, wage calculation, payment records, overtime working and worker performance while improving accuracy, transparency, accountability, efficiency and accessibility of labour related information.

2.2 Sub Objective

1. Worker Information Management: To implement system for registering and managing the worker information and employment details.
2. Attendance and Work Record Management: To record and manage the attendance and working hours along with other necessary work details needed to calculate wages accurately.
3. Wage Calculation Management: To automate the process of calculating worker wages from the attendance and working hours records as well as overtime and any deduction.
4. Payment Management: To keep track of payment record which include wage payment date, wage amount and whether it has been paid or not.
5. Labour Cost Monitoring: To allow management team to get access to accurate labour related and wages details to control labour cost
6. Improved Data Accessibility and Transparency:
7. Reporting and Decision Support:

3.Stakeholder Identification

Table 3.1 Stakeholder Identification

Stakeholder	Type (Primary/Secondary/Tertiary)	Interest/Concern
HR Manager	Primary	Need accuracy in labour records, record keeping efficiency, duplicate elimination and skill availability information
Project Manager	Primary	Need accurate information about workers to approve wage statement to processed verified payments
Site Supervisor	Primary	Need efficient way to record workers attendance, working hours and overtime
Payroll Officer	Primary	Need accurate worker attendance, overtime wage information to calculate wage payment correctly
Finance Officer	Secondary	Need approved and accurate payment information to process payments and record it correctly
Construction Worker	Primary	Want accurate and personal work records, suitable work records and correct timely wage payment

4.Procedure

The project schedule is presented below. The complete and editable Gantt chart can be accessed using the link provided.

[J A L Enterprises Construction Labour Management System.mpp](#)

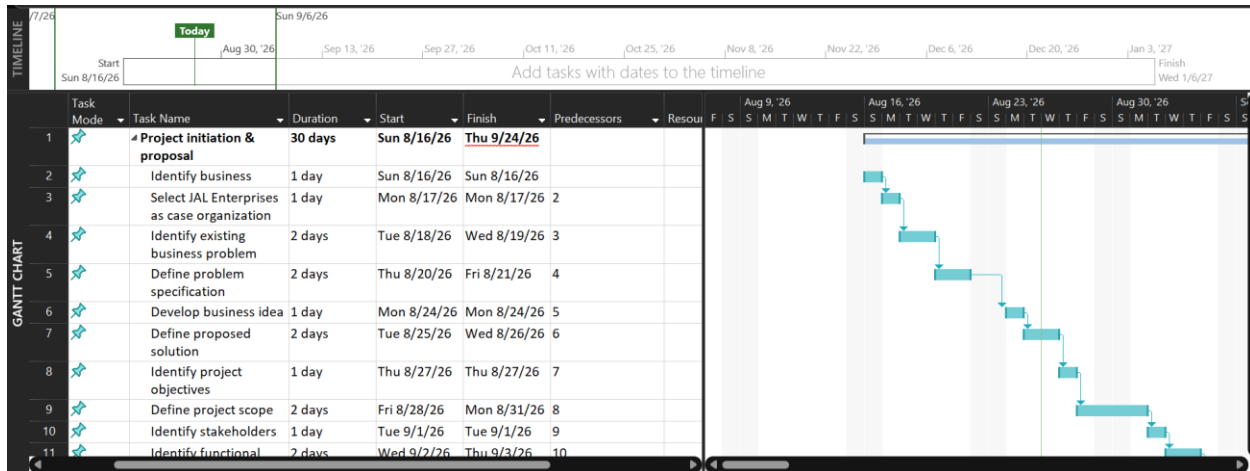


Figure 4.1 Gantt Chart of the Web Based Construction Management System

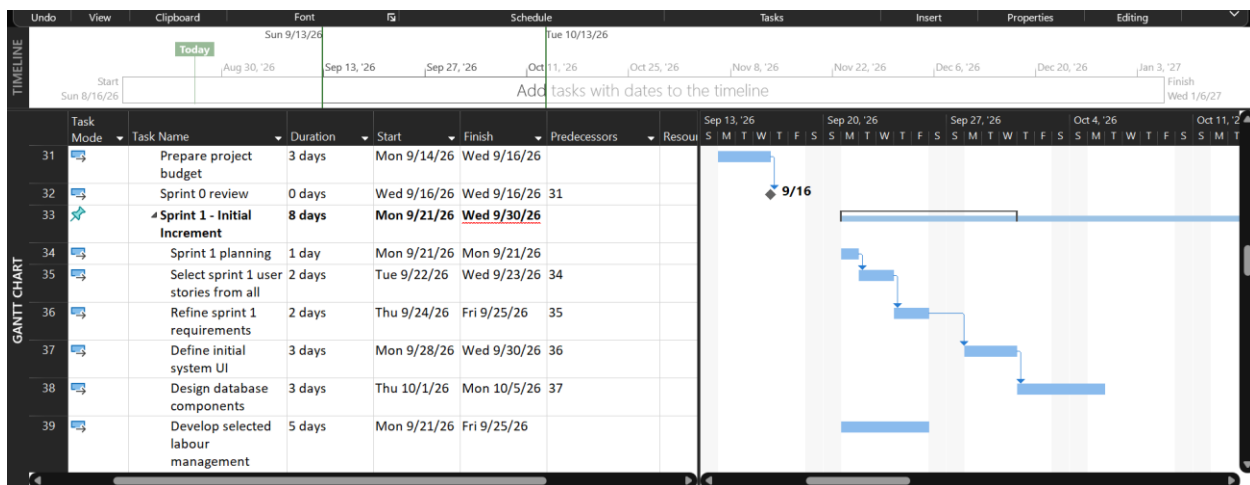


Figure 4.2 Gantt Chart of the Web Based Construction Management System (2)

5.Product Vision Statement

To the construction firms which is suffer form delay, mistakes and lack of organized records in the management of laborers, our Construction Labour Management System us the answer to all your problems in terms of registering labourers, allocating sites, tracking attendance, overtime and payment.

6.Epics

Table 6.1 Epics

Epics and Epics ID	Epic Description
Epic 1 - Labour Management	Manages the process of maintaining and registering information of construction workers, such as the worker's profile, personal information, employment information and job profile. It gives us the centralized source of information to manage the construction labor force.
Epic 2 - Site Allocation and Work Assignment	Manage worker assignment to construction site and task assignment to those workers. It allows authorized users to manage worker assignment according to project needs and check assigned tasks.
Epic 3 – Attendance and Overtime Management	Manages daily workers attendance, working hours, overtime and related attendance records. This provides necessary data for worker wage.
Epic 4 - Wage and Payment Management	Manages the calculation and processing of worker wage based on their attendance, working hours and overtime working hours. It also maintain the payment records, payment status, payment history to improve labour payments and their dedication for the work.

7. User stories

Table 7.1 User stories

Epic ID	User story ID	User story	Stakeholder
Epic 1	US 1.1	As an HR/Administration staff member, I want to register a new labourer with their personal, identification, contact, skill and wage-related information, so that a complete labourer profile is available for future site allocation and payroll.	HR/Administration
Epic 1	US 1.2	As an HR/Administration staff member, I want the system to validate required labourer information during registration, so that incomplete records are identified before they are used by other departments.	HR/Administration
Epic 1	US 1.3	As an HR/Administration staff member, I want to search for and identify possible duplicate labourer records, so that duplicate profiles are not unnecessarily created.	HR/Administration
Epic 1	US 1.4	As an HR/Administration staff member, I want to update a laborer's profile when their personal, contact, skill or wage related information changes, so that the organization maintains accurate and up-to-date records.	HR/Administration
Epic 1	US 1.5	As a Site Manager, I want to view relevant labourer skill and availability information, so that I can	Site Manager

		make informed decisions when planning labour allocation.	
Epic 2	US 2.1	As a Project/Site Manager, I want to record the labour requirement for a construction site or project activity, so that the organization can identify the number and type of labourers needed.	Project/Site Manager
Epic 2	US 2.2	As an HR/Administration staff member, I want to search labourer records based on skills and availability, so that suitable labourers can be identified for a site or task.	HR/Administration
Epic 2	US 2.3	As a Project/Site Manager, I want to review and approve suitable labourer allocations, so that labourers are assigned to appropriate construction sites and work activities.	Project/Site Manager
Epic 2	US 2.4	As a Site Supervisor, I want to receive and confirm labourer assignment information, so that I can verify whether the allocation is practical and confirm the labourer's arrival at the site.	Site Supervisor
Epic 2	US 2.5	As a Project/Site Manager, I want to revise an assignment when a labourer is unavailable or the allocation is impractical, so that suitable labour coverage can be maintained with minimum delay.	Project/Site Manager
Epic 3	US 3.1	As a Site Supervisor, I want to mark a labourer's daily attendance and record their working hours, so that accurate work-time information is	Site Supervisor

		maintained for operational monitoring and wage calculation.	
Epic 3	US 3.2	As a Site Supervisor, I want to record and authorize a labourer's overtime hours, so that only approved overtime is included in the wage calculation.	Site Supervisor
Epic 3	US 3.3	As an HR/Administration staff member, I want to review attendance and overtime records for missing or unclear entries, so that incomplete information can be corrected before payroll processing.	HR/Administration
Epic 3	US 3.4	As an HR/Administration staff member, I want to consolidate verified attendance and overtime records from construction sites, so that a reliable record of labour hours is available for payroll.	HR/Administration
Epic 3	US 3.5	As an Wage/Payroll staff member, I want to access verified attendance and approved overtime hours, so that accurate working hours can be used for wage calculation.	Wage/Payroll Officer
Epic 4	US 4.1	As a Wage/Payroll Officer, I want to collect and verify workers' wage, attendance and overtime information, so that accurate information is available for wage calculation.	Wage/Payroll Officer
Epic 4	US 4.2	As a Accounts/Payroll Officer, I want to calculate a worker's regular wages and overtime payments based on verified work records, so that the correct amount is prepared for payment.	Accounts/Payroll Officer

Epic 4	US 4.3	As a Project Manager, I want to review and approve calculated wages and payment requests, so that payments are authorized before they are processed.	Project Manager
Epic 4	US 4.4	As a Finance/Accounts Officer, I want to process and record approved worker payments, so that completed payment transactions are properly documented.	Finance/Accounts Officer
Epic 4	US 4.5	As a Wage/Payroll Officer, I want to view and maintain wage and payment records, so that accurate payment history is available for future reference and verification.	Wage/Payroll Officer

8.Initial Product Backlog

Table 8.1 Initial Product Backlog

User story ID	Remarks	Estimated story points	Estimated Hours	Priority (High/medium/low)
US 1.1	Core function for creating labourer records.	5	10	High
US 1.2	Prevents incomplete records from entering the system.	3	6	High
US 1.3	Reduces duplicate profiles and	5	10	Medium

	inaccurate records.			
US 1.4	Supports maintenance of accurate labourer profiles.	3	6	High
US 1.5	Supports site allocation and planning.	3	6	High
US 2.1	Captures the demand that starts the allocation process.	3	6	High
US 2.2	Helps identify suitable workers for a requirement.	5	10	High
US 2.3	Confirms the proposed site/task assignment.	5	10	High
US 2.4	Allows the supervisor to verify that the allocation is practical.	3	6	Medium
US 2.5	Handles allocation problems and labour shortages.	5	10	Medium
US 3.1	Essential source of verified	5	10	High

	work-time information.			
US 3.2	Provides approved overtime information for payroll.	5	10	High
US 3.3	Ensures records are corrected before payroll.	3	6	High
US 3.4	Creates a reliable consolidated record for payroll.	5	10	High
US 3.5	Provides the required input to the payment process.	3	6	High
US 4.1	Ensures required inputs are available before calculation.	3	6	High
US 4.2	Core payroll calculation functionality.	8	16	High
US 4.3	Provides management authorization before payment.	5	10	High
US 4.4	Completes the payment process and creates a	5	10	High

	transaction record.			
US 4.5	Supports payment history, verification and record keeping.	3	6	Medium

9. Definition of Done (DOD)

A user story is “Done” when the functionality required has been implemented in accordance with the acceptance criteria, has been tested successfully, has been integrated with the relevant system components and has been reviewed by the development team. Functionality should work without critical errors. Functionality should meet defined business requirements. Required documentation should be completed. The functionality has to be accepted by the Product Owner/stakeholder as done.

A user story will be considered done when;

- The functionality as described in the user story is implemented.
- Meets all acceptance criteria.
- The functionality has been successfully tested.
- No critical or high severity issues remaining.
- The functionality works as expected with the system components concerned.
- The relevant database operations are working correctly where applicable.
- The user interface is usable and functions as expected.
- The development team has reviewed the coding/functionality.
- Updated user or technical documentation as required.
- The story has been reviewed and accepted by the appropriate stakeholder or Product Owner.

10.Sprint 1 Preview

Epic 1:

1. US 1.1

- Add a new laborer to the system.
- Capture details such as Name, Contact Details, NIC/Employee ID, Job Role and other required information.
- Validation of required fields.

2. US 1.2

- Enable authorized users to search and view laborers information.
- Clearly show the relevant employee information.

3. US 1.3

- Allow authorized users to edit existing laborer information.
- Save the updated information properly.

Epic 2 :

1. US 2.1

- Add a new construction site/project.
- Capture site name, location, project details and dates relevant to the project

2. US 2.2

- View registered construction sites.
- Search for a particular site when required.

3. US 2.3

- Update site information when project details change.
- Ensure the new information is stored properly.

Epic 3 :

1. US 3.1

- Records attendance status such as Present/Absent
- Stores the attendance records

2. US 3.2

- Supervisor can view attendance records
- Search/ filter by labourer, site or date

3. US 3.3

- Correct an attendance record if an error was made
- Maintain the updated records

Epic 4 :

1. US 4.1

- Use the attendance and overtime data to determine the basic wage.
- Use the appropriate wage rate.
- Include overtime where applicable.

2. US 4.2

- Authorized users should be allowed to view the calculated wages.
- Show the regular hours, the overtime hours, the rates, and the total wage.

3. US 4.3

- Record payment made to the labourer.
- Save the payment date, amount, and other important payment details.

11. Software and Hardware Requirement

Table 11.1 Hardware requirements

Component	Minimum Requirement
Development computer/Laptop	Intel Core i5, 8 GB RAM, 256 GB SSD, Windows 10/11 or equivalent
Processor	Dual core 2.0 GHz or Higher
RAM	Minimum 8 GB
Storage	Minimum 256 GB SSD with sufficient free space for development tools and database
Display	1366 x 768 resolution or higher
Network Connection	Stable WiFi/Ethernet connection, minimum 10 Mbps
Server	Minimum 2 CPU cores, 4 GB RAM, 50 GB storage for small scale deployment

Table 11.2 Software components

Category	Technology/ tool	Purpose
Operating system	Windows 10/11	Development and system operation
Frontend Development	HTML5, CSS3, JavaScript	Creating the user interface
Frontend Framework	React.js	Building interactive and responsive web interfaces
Backend Development	Node.js	Running the server side application
Backend Framework	Express.js	Developing APIs and handling backend requests

Database	MySQL	Storing employee, attendance, overtime, wage and payment data
Version control	Git & GitHub	Tracking changes and managing source code versions
Repository	GitHub	Hosting and collaborating on the project source code

To carry out development, testing, deployment, and maintenance the system needs only basic computer hardware and some commonly used software tools. The hardware required comprises development computers, a server, network connectivity and mobile devices for testing purposes. The software includes the frontend and backend development technologies, a database management system, coding and version-control tools, API testing tools, and design and documentation applications. These resources are enough to enable efficient development and operation of the system.

12. Budget

Table 12.1 Budget

Budget Item	Estimated cost (LKR)
Development Laptop/computer (Existing team computer/laptop and devices used for development and testing.)	0
Server/ cloud hosting	30,000
Internet and networking	10,000
Domain name	5,000
Database and development software	0
UI/UX and diagramming tools	0
Testing devices and miscellaneous equipment	15,000
Printing and documentation	5,000
Contingency	20,000
Estimated total	85,000

The estimated project budget is LKR 85,000, covering the main resources required for developing, testing, and deploying the system. The largest expense is hosting while internet, testing, documentation, and other project-related costs make up the remaining budget. Most development and design software can be obtained free of charge or through open-source solutions, helping to keep the overall project cost relatively low. A contingency amount is also included to cover unexpected expenses.

13. References

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